All-order monotonicity of iterated parabolic-cylinder ratios

MathDB 364320: a spectral proof

Principia Math In Progress

17 September 2026; source audit updated 22 September 2026

Task 1: the original problem. For $n > 1/2$ and $x \in \mathbb{R}$, set

$$R_n^{[1]}(x) = \frac{U(n-1,x)}{U(n,x)}, \qquad R_n^{[k+1]}(x) = \frac{R_n^{[k]}(x)}{R_{n+1}^{[k]}(x)}.$$

Here U is the standard parabolic cylinder function. We prove the substantive all-order assertions of Segura's Conjecture 3.6 in [1]. The comparison between successive orders starts at $k = 2$: the unqualified comparison printed in the source cannot include $k = 1$, since $R_n^{[1]}(x) \sim x$ and $R_n^{[2]}(x) \to 1$ as $x \to +\infty$. We do not use that indexing issue as a counterexample or as research credit.

Theorem 1. *For every real $n > 1/2$, every real x, and every integer $k \geq 1$, $R_n^{[k]}(x) > 0$ and*

$$\frac{d}{dx} R_n^{[k]}(x) > 0.$$

For every integer $k \geq 2$,

$$0 < R_n^{[k]}(x) < R_n^{[k+1]}(x) < 1, \qquad \lim_{x \to +\infty} R_n^{[k]}(x) = 1.$$

Furthermore, for every fixed $n > 1/2$ and finite real x,

$$\lim_{k \to \infty} R_n^{[k]}(x) = 1.$$

Publication scope. The proof below is complete for these quantifiers. All 23 pages of the author's manuscript announced as forthcoming in AMS Contemporary Mathematics have now been read; Conjecture 3.6 remains unresolved there. Acceptance is inferred from the explicit author announcement, supported by the reviewer acknowledgement and publication list. The audit assigns one new campaign credit on that source basis. The separate typeset publisher chapter remains unread; no identity with it is asserted. The accompanying audit records this revision of the earlier stricter publisher-copy requirement and the bounded public-claim checks. The mathematical proof is unchanged.

A positive spectral representation

Write $D_{-a}(t) = U(a - 1/2, t)$. For $a > 0$,

$$D_{-a}(t) = \frac{e^{-t^2/4}}{\Gamma(a)} \int_0^\infty s^{a-1} e^{-s^2/2 - ts}\, ds > 0, \qquad D_0(t) = e^{-t^2/4}. \tag{1}$$

In particular all ratios in the theorem are defined and positive. The integral also gives Gaussian decay, with at most polynomial factors, as $t \to +\infty$, including the first two t-derivatives. The standard equation and derivative identity are

$$D''_{-a}(t) = \left(\frac{t^2}{4} - \frac{1}{2} + a\right) D_{-a}(t), \qquad D'_\nu(t) = \frac{t}{2} D_\nu(t) - D_{\nu+1}(t). \tag{2}$$

They can also be obtained by differentiating (1) and integrating by parts; see [2] for the normalizations.

Fix a real endpoint x. Let H_x be the Dirichlet self-adjoint realization on $L^2(x,\infty)$ of

$$-\frac{d^2}{dt^2} + \frac{t^2}{4} - \frac{1}{2}.$$

More explicitly it is the Friedrichs operator of the closure, initially on $C_c^\infty(x,\infty)$, of the quadratic form

$$\int_x^\infty \left(|u'|^2 + \left(\frac{t^2}{4} - \frac{1}{2}\right) |u|^2 \right) dt = \int_x^\infty \left| u' + \frac{t}{2} u \right|^2 dt. \tag{3}$$

Thus $H_x \geq 0$. It has no zero eigenvector: equality in (3) would imply $u(t) = ce^{-t^2/4}$, and its Dirichlet boundary value forces $c = 0$.

Put

$$v_a(t) = \frac{D_{-a}(t)}{D_{-a}(x)}, \qquad v_0(t) = e^{-(t^2-x^2)/4}, \qquad m_x(a) = -\frac{D'_{-a}(x)}{D_{-a}(x)}.$$

The function v_0 is a nonzero vector of $L^2(x,\infty)$; it is not asserted to satisfy the Dirichlet boundary condition. Let E_x be the spectral resolution of H_x and define the finite positive measure

$$\nu_x(B) = \langle v_0, E_x(B) v_0 \rangle.$$

It has positive total mass and is supported on $(0,\infty)$, since $H_x \geq 0$ and $\ker H_x = \{0\}$.

Lemma 1. *For every $a > 0$,*

$$m_x(a) = \frac{x}{2} + \int_{(0,\infty)} \frac{a\lambda}{\lambda + a} \, d\nu_x(\lambda). \tag{4}$$

Consequently,

$$m_x(a+1) - m_x(a) = \int_{(0,\infty)} \frac{\lambda^2}{(\lambda + a)(\lambda + a + 1)} \, d\nu_x(\lambda). \tag{5}$$

Proof. Both v_a and v_0 have boundary value 1. Their difference has zero boundary value, is square integrable, and, by (2) and the decay stated above, belongs to the Dirichlet operator domain. The equation for this difference is

$$(H_x + a)(v_a - v_0) = -a v_0.$$

The resolvent exists for $a > 0$, so

$$v_a = v_0 - a(H_x + a)^{-1} v_0. \tag{6}$$

On the other hand, the Wronskian $W(t) = v_0(t) v'_a(t) - v'_0(t) v_a(t)$ satisfies $W'(t) = a v_0(t) v_a(t)$. Its limit at infinity is zero, and $W(x) = -m_x(a) + x/2$. Integration therefore gives

$$m_x(a) - \frac{x}{2} = a\langle v_0, v_a \rangle = a\langle v_0, [I - a(H_x + a)^{-1}] v_0 \rangle.$$

The spectral theorem and (6) give (4). Its integral is finite because $0 \leq a\lambda/(\lambda + a) \leq a$ and ν_x is finite. Subtracting the two representations proves (5); its integrand lies between zero and one. □

An explicit positive formula at every order

Set $a = n - 1/2 > 0$. Since $R_n^{[1]} = D_{-a}/D_{-a-1}$, (5) gives

$$\frac{d}{dx} \log R_n^{[1]}(x) = m_x(a+1) - m_x(a) = \int \frac{\lambda^2}{(\lambda+a)(\lambda+a+1)}\, d\nu_x(\lambda). \tag{7}$$

The measure is held fixed while taking differences in a. No derivative of the endpoint-dependent measure with respect to x is being taken.

Let $\Delta f(a) = f(a+1) - f(a)$ and $(b)_j = b(b+1)\cdots(b+j-1)$. The defining ratio recurrence implies

$$\log R_n^{[k]}(x) = (-1)^{k-1} \Delta^{k-1} \log R_n^{[1]}(x).$$

For $b > 0$ and $p \geq 0$, the elementary identity

$$(-1)^p \Delta^p \frac{1}{b(b+1)} = \frac{(p+1)!}{(b)_{p+2}} \tag{8}$$

follows by induction from $1/(b)_j - 1/(b+1)_j = j/(b)_{j+1}$. Taking this finite linear combination in (7) yields

$$\boxed{\frac{d}{dx} \log R_n^{[k]}(x) = k! \int_{(0,\infty)} \frac{\lambda^2}{(\lambda+a)_{k+1}}\, d\nu_x(\lambda) > 0.} \tag{9}$$

All integrals converge: the kernel for $k = 1$ is bounded by one, and the ratio of the kernel for $k+1$ to that for k is

$$\frac{k+1}{\lambda+a+k+1} < 1. \tag{10}$$

Strict positivity follows from the nonzero positive measure on $(0, \infty)$. Since the ratios themselves are positive, (9) proves their strict increase in x for every $k \geq 1$.

The upper bound and both limits

In (1), substitute $s = u/x$ for $x > 0$. Dominated convergence, with majorant $u^{a-1}e^{-u}$, proves

$$D_{-a}(x) \sim e^{-x^2/4} x^{-a}, \qquad R_n^{[1]}(x) \sim x \qquad (x \to +\infty).$$

For every fixed $k \geq 2$, the expression for $\log R_n^{[k]}$ is a finite difference of $\log x + o(1)$, so its limit is zero. Strict increase in x then gives $\log R_n^{[k]}(x) < 0$ for every finite x, proving $R_n^{[k]}(x) < 1$. The recurrence immediately gives

$$R_n^{[k+1]}(x) = R_n^{[k]}(x)/R_{n+1}^{[k]}(x) > R_n^{[k]}(x) \qquad (k \geq 2).$$

For completeness, the limit in the order also follows from the same kernel. For fixed $b = \lambda + a > 0$, $k!/(b)_{k+1} \to 0$: by (10), its successive ratios are $(k+1)/(b+k+1)$, whose product tends to zero because $\sum_k \log(1 + b/(k+1)) = \infty$. Dominated convergence in (9) shows, for each fixed real t,

$$q_k(t) := \frac{d}{dt} \log R_n^{[k]}(t) \longrightarrow 0, \qquad 0 < q_k(t) \leq q_2(t) \quad (k \geq 2).$$

Moreover, the already proved limit at $+\infty$ gives

$$-\log R_n^{[k]}(x) = \int_x^\infty q_k(t)\, dt, \qquad \int_x^\infty q_2(t)\, dt = -\log R_n^{[2]}(x) < \infty.$$

A second application of dominated convergence proves $\log R_n^{[k]}(x) \to 0$. This completes the proof of Theorem 1.

Verification scope. This is an analytic proof for every required order and parameter. The accompanying symbolic checker verifies the rational kernel identities used in the proof; numerical checks verify conventions and offsets only. Neither finite computation is used to infer an infinite sign assertion. The audit was performed by the authoring agent, with no independent peer review claimed.

Four follow-up tasks

Task 2. Determine the sharp asymptotic size of $1 - R_n^{[k]}(x)$ as $k \to \infty$ in terms of the lowest Dirichlet eigenvalue of H_x, including uniform estimates as $n \downarrow 1/2$ and $x \to -\infty$.

Task 3. Compute explicit spectral weights for ν_x, including closed formulas at $x = 0$, and turn truncated versions of (9) into certified two-sided bounds with effective tail estimates.

Task 4. Identify other contiguous special-function ratios admitting the same positive-resolvent argument, specifying the parameter ranges in which the associated Dirichlet operator is nonnegative and the required ratio identities hold.

Task 5. Determine whether each $R_n^{[k]}(x)$ with $k \geq 2$ has exactly one inflection point, and how that point depends on n and k.

References

[1] J. Segura, *On bounds for ratios of contiguous hypergeometric functions*, arXiv:2408.05573v1 (2024), Conjecture 3.6, p. 9. All 23 manuscript pages were reviewed. Published in *Contemporary Mathematics* **818** (2025), 71–93, DOI 10.1090/conm/818/16368. Published Conjecture 3.6 is on p. 79; only searchable excerpts of the published chapter were accessible.

[2] NIST Digital Library of Mathematical Functions, §12.5 and §12.8, integral, recurrence, and derivative conventions for parabolic cylinder functions.